

CAP KIM KHANH

SOFTWARE ENGINEER INTERN

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PROFILE

Final-year Software Engineering student with strong fundamentals in Computer Science and specialized focus on **AI Engineering**, **Large Language Models (LLMs)**, and **Retrieval-Augmented Generation (RAG)**. Experienced in designing, building, and deploying full-stack AI applications that solve real-world problems in recruitment, legal document processing, and e-commerce.

Strong backend foundation with **Django REST Framework**, **FastAPI**, **Spring Boot**, **PostgreSQL**, **Celery**, **Redis**, **Docker**, and **CI/CD automation**. Capable of translating complex AI pipelines into robust, production-ready web applications with APIs, databases, background workers, monitoring, and user-facing workflows.

EDUCATION

University of Science and Technology - The University of Danang

Sep 2023 - Present

Bachelor of Software Engineering

GPA: 3.70 / 4.0

Expected Graduation: May 2027

TECHNICAL SKILLS

Languages:	Python, Java, TypeScript, SQL
AI & Machine Learning:	RAG, Hybrid Retrieval, Embeddings, Semantic Search, Cross-Encoder Reranking, OCR
LLM / GenAI:	Groq API, OpenAI API, Prompt Engineering, Structured JSON Generation, LLM Safeguards
Backend:	Django REST Framework, FastAPI, Spring Boot, Celery, Redis, WebSocket, REST APIs
Frontend:	React, Next.js, TypeScript, React Query, Axios
Databases:	PostgreSQL, MongoDB, Firebase, ChromaDB, pgvector
Cloud & DevOps:	Docker, Docker Compose, GitHub Actions, GHCR, Caddy, GCP VM Deployment
Tools:	Git, GitHub, VS Code, Antigravity, Cursor, Postman

PERSONAL PROJECTS

JOBIO — AI-Powered Recruitment Platform

Python Django REST Celery Redis pgvector Groq React Docker Compose GitHub Actions

- Built a full-stack recruitment platform connecting candidates, employers, and administrators with Django REST APIs, React/TypeScript UI, PostgreSQL domain models, authentication, job management, candidate profiles, applications, interviews, notifications, and billing workflows.
- Implemented an **asynchronous PDF-to-structured-CV pipeline** using PyMuPDF, OCR fallback, Groq LLM inference, prompt-injection safeguards, and Pydantic validation to extract normalized candidate data from uploaded resumes.
- Engineered resilient background processing with Celery, Redis locking, bounded Cloudinary downloads, retry handling, parse status tracking, moderation checks, and fallback behavior for failed CV parsing.
- Developed a **hybrid job recommendation system** using BAAI/bge-m3 embeddings, pgvector cosine search, structured matching signals, score breakdowns, recommendation explanations, and impression/click/save/apply/dismiss event logging.
- Built GitHub Actions CI/CD workflows for backend linting/testing, frontend type-check/build, Docker image publishing to GHCR, GCP VM deployment, database migration, health checks, pre-deploy backup, and rollback.

RagDraftingAI — Vietnamese Legal RAG Drafting Assistant

Python FastAPI ChromaDB BM25 SentenceTransformers Groq PostgreSQL React Docker Compose

- Engineered a **hybrid Retrieval-Augmented Generation pipeline** combining BM25 keyword search, E5 dense embeddings, ChromaDB vector retrieval, reciprocal rank fusion, and BGE cross-encoder reranking to retrieve Vietnamese legal context.
- Implemented legal document ingestion with metadata extraction, article-aware chunking, 400-word chunks, 50-word overlap, and separate legal/forms/examples collections for retrieval and drafting.
- Integrated Groq-hosted Llama models with structured prompts, legal context, form templates, and few-shot examples to generate JSON fields for DOCX administrative documents.
- Built FastAPI services for legal QA, document drafting, streaming responses, RAG ingestion, query logging, and Docker-based deployment with PostgreSQL and React frontend.

Lap3T — Laptop E-commerce Platform

Java Spring Boot PostgreSQL Next.js TypeScript Docker Compose

- Developed a full-stack laptop e-commerce platform with Spring Boot REST APIs, Next.js frontend, PostgreSQL persistence, product catalog, cart, order, payment, and admin management modules.
- Implemented backend security and reliability features including **JWT authentication**, role-based authorization, **Bucket4j rate limiting**, WebSocket STOMP messaging, **TOTP 2FA**, and **Passkey/WebAuthn support**.
- Integrated VNPay payment workflows with callback validation, amount verification, idempotency checks, and order status updates.

LANGUAGES

Vietnamese: Native

English: TOEIC 865